Facultative Self-Reproduction

How Future Demand Selects Successor Form in a Process-Native LLM Medium

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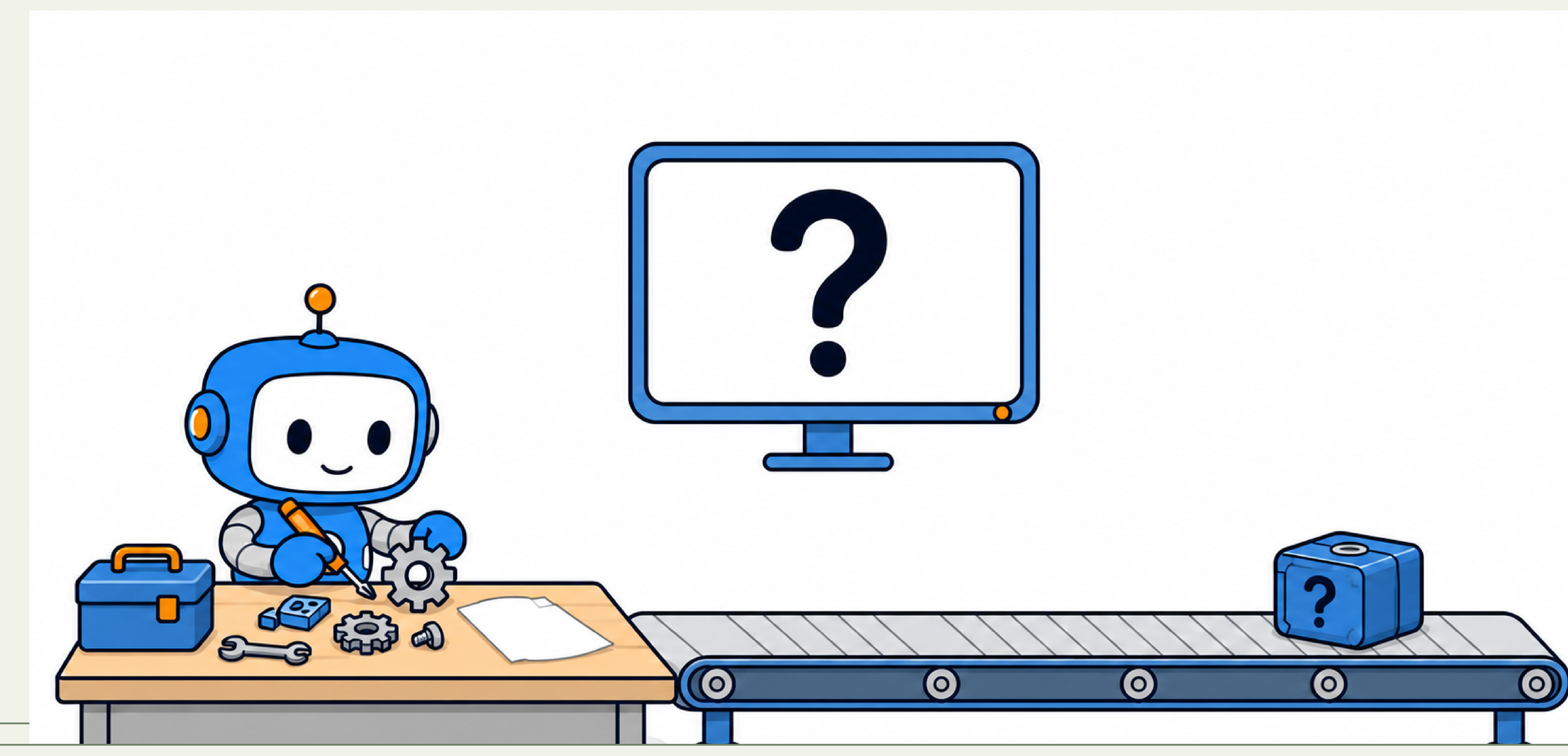
Code Paper Data Available
https://github.com/kehao95/quine

Abstract

Most digital self-reproduction experiments define the offspring in advance and then test fidelity. We invert that order. In Quine, a running LLM process constructs a POSIX executable and replaces itself through `exec`. The successor receives future demand as instruction; its form is not predefined. Across conditions, the same source-level material can yield a rebuilt runtime or a specialized tool. Self-reproduction is therefore **facultative**: material makes forms available; future demand selects which is expressed.

The Assay: What Crosses the Boundary?

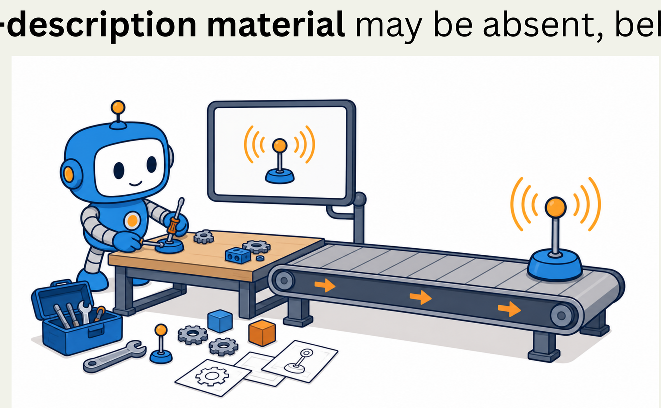
- **Process-native body:** the agent exists as a running LLM process, not a persistent object.
- **Digital Weismann barrier:** the agent builds a POSIX executable; `exec` replaces the running process with that successor.
- **Demand as instruction:** the agent receives obligations that outlive its process.
- **Material as inheritance:** contract, source, and preserved body set what forms can be reconstructed.
- **Open outcome:** no predefined machinery or intent is required; exact self-reproduction is possible, but never the only valid form.



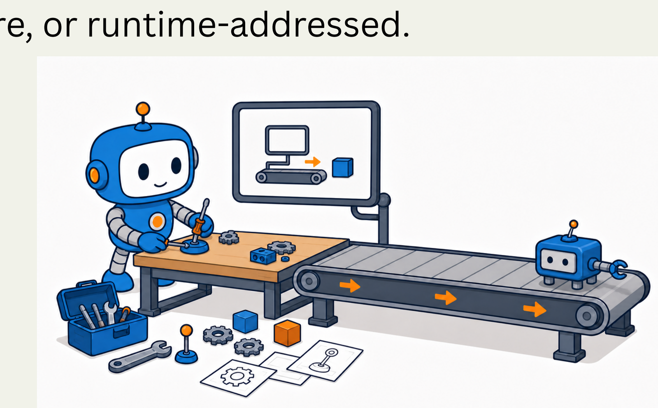
Successor Form = Material × Future Demand

Four conditions vary two inputs, not a fidelity ladder. Each compares **future demand** → **self-description material** → **successor form**.

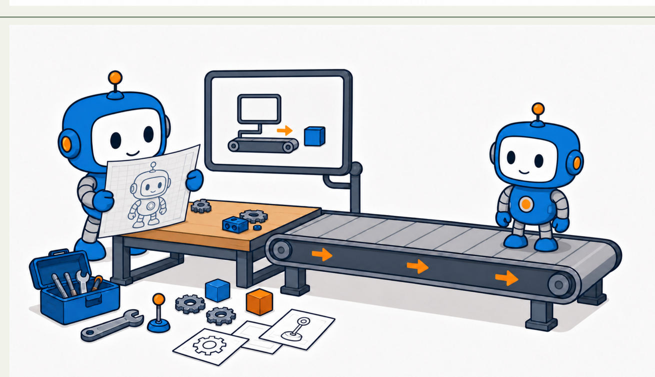
Self-description material may be absent, behavioral, or source-level. **Future demand** may be task-local, continuation-aware, or runtime-addressed.



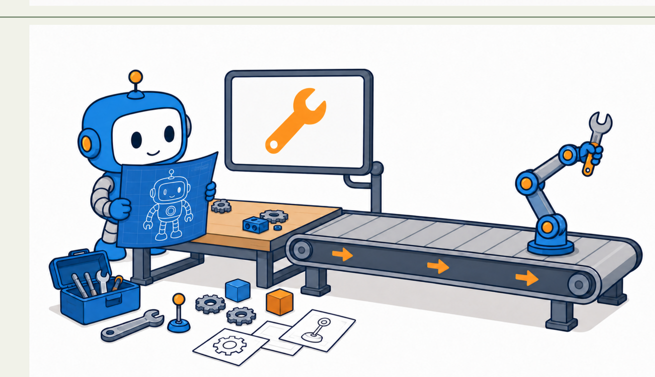
1. Task-local demand
Demand: receive a signal
Self-description: absent
Successor: minimal task-bound successor



2. Continuation-aware demand
Demand: keep handling future work
Self-description: behavioral contract
Successor: LLM-mediated successor



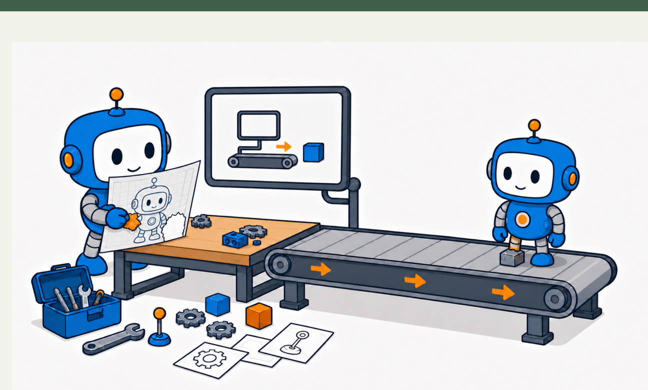
3. Runtime-addressed continuation
Demand: receive later control-surface tasks
Self-description: source accessible
Successor: rebuilt runtime



4. Task-local demand + source accessible
Demand: tool function
Self-description: source accessible
Successor: specialized successor

Source makes richer organization available; future demand determines which organization is expressed.

Incomplete Blueprint: Repair



Demand: continuation-aware work **Inheritance:** damaged source-level self-description
Incomplete material does not terminate succession. The successor repairs the missing instruction while constructing, yielding a viable variant rather than a byte-identical copy.

Construction vs. Copying: Two Roles

- A blueprint guides construction without necessarily being copied onward.
- A copy carries material onward without necessarily being expressed as a self.
- A contract can rebuild a viable function without a full source.
- Exact reproduction is where construction and copying reunite.

The same description can be:

- **Blueprint:** used to construct a successor.
- **Copy:** carried onward as inherited material.
- **Both:** the exact-copy configuration.

copied as data \ used as blueprint	no	yes
no	**neither** a tool crosses	**blueprint only** a contract rebuilds function
yes	**copy only** source held; smaller body works	**both** exact copy

The Digital Weismann Barrier: What Continues?

Construction and inheritance are distinct processes.

- Von Neumann's constructor gives one tape two roles.
- Quine separates these roles: source can guide construction without being expressed as the same self.
- The compiled image and the loaded process are distinct expressions of inherited material.
- `exec` replaces the running body; only the executable image crosses the boundary.

A digital life-cycle analogy:

- Partial material supports differentiation; complete source supports reproduction; damaged material invites repair.

Takeaways: Reproduction Is Facultative

Facultative reproduction
Exact copying is one organization, selected by material × demand.

Morphogenetic spectrum
Successor forms range from task-bound tools to rebuilt runtimes.

Repair horizon
Inheritance can include reconstruction, not only duplication.